Proof of Stake via Partial Common Ownership

A Public Goods Property Mechanism for Decentralised Validator Networks

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Executive Summary

This paper is a first iteration toward solving a problem that has frustrated every attempt to build fair, stable decentralised networks: how do you reward the people who do the work without letting speculators price them out? We do not claim to have the final answer. We present an approach, a set of tools, and our early thinking — honestly including what we have shown and what remains to be proven.

The answer draws on a surprisingly old idea. In the 1870s the economist Henry George observed that land speculators — people who buy property and hold it idle, waiting for prices to rise — extract value from a community without contributing to it. His solution was a tax on the land itself, not on what is built on it. The tax made idle holding costly, which forced land into productive use, which benefited everyone.

We apply exactly this logic to validator positions in a decentralised network. A validator is someone who runs a node providing a service — in our case, coordinating federated AI model training on the InfiniteZero Network. We want validators who show up and do the work. We do not want speculators who buy validator positions and hold them idle, hoping to sell at a profit.

The proposed mechanism in one paragraph

Every validator holds an NFT that functions like a property deed. The DAO — the network's governing body — acts like a land registry. Validators pay a small continuous tax simply for occupying their position in the network. They also share a portion of their earnings with the DAO when the network is active, with longer-serving validators paying a lower rate as a reward for loyalty. In return, the DAO guarantees a buy-back payment at the end of the validator's ten-year term — a guaranteed exit price that rises with each year of honest service. If the network earns nothing for years at a time, the earnings share drops to zero automatically. The small site tax continues, because occupying a position in the network is still worth something even when the network is quiet.

What this paper establishes so far

- **A solvency condition we believe is provable.** Under certain conditions, we believe the DAO can honour every buy-back promise even if the network earns zero revenue for the entire ten-year lease. We have derived a candidate closed-form condition for this and proven it within the model's assumptions.
- **A mechanism we believe structurally discourages speculation.** We believe the continuous site tax makes idle speculative holding costly enough to discourage it. Whether this holds under all realistic conditions is one of the open questions.
- **An incentive structure intended to reward long-serving validators.** The seniority system is designed so that validators who have served longest earn more and pay less. Tenure follows the staking record rather than the holder, which we intend to prevent gaming — though the formal verification of this is part of the ongoing work.
- **The network handles every scenario.** A boundary condition treatment that attempts to specify the mechanism's behaviour from complete dormancy to explosive growth — though some of these responses remain to be formally verified.

What remains open

Four theoretical questions are not yet resolved — and we think that is worth being upfront about. We describe them honestly in Section XII. Each is a genuine open research problem, not a minor detail. Whether the mechanism ultimately holds up under rigorous scrutiny in these areas is part of what we are seeking to find out. The companion hiring materials describe the research programme to address them.

Abstract

We present a complete mechanism design for decentralised validator networks that simultaneously achieves security through staked property rights, access through anti-speculative pricing, and accountability through enforceable slashing. The mechanism synthesises Georgist land value theory, Harberger partial common ownership, and proof-of-stake security into a single programmable framework on Ethereum.

The central contribution is the decomposition of validator NFT value into a site component (DAO-set, continuously taxed as Georgist ground rent, anti-speculative) and a service component (self-assessed, subject to Harberger forced-sale, zero cost during dormancy). This decomposition resolves the classical tension between anti-speculation and idle-holding cost that no prior mechanism has achieved.

Four theorems are proven: unconditional solvency (Theorem 1), anti-oligopoly bounds (Theorem 2), fixed-point existence via Kakutani's theorem (Theorem 3), and generational floor boundedness (Theorem 4). A complete boundary condition treatment covers total collapse through infinite growth. Four open theoretical problems are honestly stated. The mechanism requires three protocol fill-ins (κ, p_unit, F_0_min) and is otherwise fully specified.

I. Introduction

Every decentralised network that requires validators — nodes that do real work to keep the system running — faces the same three-way tension.

First, validators need skin in the game. A network where validators face no cost for misbehaving or going offline is a network that will be attacked. This calls for stake: validators must put something at risk that they will lose if they act badly. Second, the people who actually want to run validators should be able to afford to participate. This means validator positions should be priced at the value of the service they provide, not at speculative premiums driven by scarcity. Third, misbehaviour must carry real, enforceable consequences — not just theoretical ones that can be renegotiated away.

These three requirements — security, access, accountability — have never been achieved simultaneously in a decentralised network. Token-based staking solves security but creates speculation: a token whose value depends on the network's perceived future will trade at prices that price out genuine validators. Pure leasing without property rights solves access but undermines security: a leaseholder with no residual claim invests less in quality. Heavy slashing solves accountability but raises entry costs so high that only well-capitalised players can participate.

This paper resolves the tension by treating validator participation as a form of property in the Georgist sense — a right to occupy a productive position in a shared commons, subject to a ground rent that prevents speculative hoarding, backed by a guaranteed exit value that rewards honest service.

The analogy: **A validator position in this network is like a plot of land in a productive district. The DAO is the state that manages the commons. The site tax is the ground rent the state charges simply for occupying the position. The earnings share is the tax on what you build on your plot. The floor price is the leaseholder's guaranteed right of renewal or buy-out at the end of the term. The Harberger forced-sale rule is the**

compulsory purchase power: if someone values your position more productively than you do, they can acquire it at your declared price.

1.1 The InfiniteZero Network

The immediate application is the InfiniteZero Network — a decentralised AI commons built on Ethereum where federated machine learning happens at the edge. Validators coordinate model training across user devices, verify encrypted model updates, and route inference requests. Their work is variable: some periods have high demand, others have none. The mechanism must incentivise validators to remain through dormancy without allowing them to profit from dormancy by speculating on position scarcity.

1.2 Running example

Throughout this paper we use a concrete example to make the mathematics tangible. We fix the following values:

- F_0 = 1,000 USDC (genesis floor price per validator NFT)
- N_0 = 20 validators at genesis
- α = 0.40 (tenure floor premium rate)
- τ_s = 3% per year (site tax rate)
- $τ\_e_0$ = 12% (base earnings share for new validators)
- λ = 0.20 (tenure earnings discount rate)
- T_lease = 10 years

All formal results are stated generally. The running example shows what the results mean in practice for a validator on the InfiniteZero Network.

II. Where the Ideas Come From

The mechanism draws on three bodies of thought that have never previously been combined. Understanding each one makes the mechanism's logic much easier to follow.

2.1 Henry George and the land tax

Henry George was a nineteenth-century economist who watched San Francisco grow from a small city into a major port and noticed something troubling: as the city became more productive, the people who owned land became wealthy not because they had done anything useful, but simply because they had been in the right place at the right time. The value of their land came from the city around them — the roads, the institutions, the other people — not from their own effort.

His proposed solution was a tax on the value of land itself — not on buildings, not on what was produced, but on the site value: what the location was worth simply by virtue

of existing in a productive community. This tax would be continuous. It would apply whether the landowner was farming, building, or doing nothing. That last part is critical: a tax that applies even when you are doing nothing makes it impossible to profit from idle waiting. Speculation becomes unprofitable. Land flows to its most productive use.

The analogy: In our mechanism: the network itself — other validators, users, the protocol infrastructure — creates the value of a validator position. An individual validator's position is worth something even before they do a single unit of work, simply because it exists in a live network. This is the 'site value' of the position. We tax it continuously, just as George proposed taxing land. A speculator who buys a validator position and does nothing pays this tax every year. Over time, the tax erodes any gain from waiting.

2.2 The Harberger mechanism

Arnold Harberger proposed an elegant solution to a different problem: how do you make sure that assets flow to whoever values them most productively, without giving up the efficiency of a market?

His answer: everyone declares the value of their assets. They pay a tax proportional to their declared value. And critically, anyone can buy the asset from them at the declared price at any time. This creates a precise incentive: if you declare too high, you pay too much tax. If you declare too low, someone buys your asset for less than it is worth to you. The only rational declaration is your true valuation.

Weyl and Posner formalised this in their 2018 book Radical Markets as the Common Ownership Self-Assessed Tax (COST). The mechanism simultaneously extracts rent for the public and ensures assets are held by whoever can use them most productively.

The analogy: In our mechanism: each validator declares the value of their position as a going concern (V^service). Any buyer can acquire it at that declared price. The validator has no incentive to declare too low (they would lose their position cheaply) or too high (the declared price is the price they will be paid if they want to sell). This is the Harberger forced-sale rule. It ensures that validator positions flow to active service providers rather than sitting with incumbents who have stopped doing useful work.

2.3 Proof of stake

The standard approach to securing decentralised networks is proof of stake: validators lock up capital (their 'stake') that can be confiscated if they misbehave. The security argument is simple — the cost of attacking the network must exceed the gain, and validators who stand to lose their stake are honest by rational self-interest.

Our mechanism replaces the fungible token stake with an NFT whose value reflects productive service. A validator's stake is the value of their validator position, which rises with honest tenure and falls with misbehaviour. This is a richer instrument than a fixed-quantity token stake, because the stake value is tied to what the network role is actually worth.

III. The Central Idea: Splitting the NFT's Value

The most important conceptual contribution of this paper is a simple decomposition. Every validator NFT has two types of value, and we treat them differently.

In plain English: **Every validator NFT has a site value (what the position is worth simply by virtue of existing in the network) and a service value (what the position is worth as a going productive concern). The site value is set by the DAO and taxed continuously. The service value is declared by the holder and governs transferability.**

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V_i  =  V_i^site  +  V_i^service
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Total NFT value = the right to exist in the network + the value of the service being provided

V^site is the Georgist component. It is not self-assessed — it is set by the DAO based on how scarce validator positions are. When there are few validators, site value is high (the position is scarce and therefore valuable to the network). When there are many validators, site value falls. The DAO taxes V^site continuously. This is what prevents speculation.

V^service is the Harberger component. It is declared by the holder. It represents what the validator genuinely believes their service contribution is worth as a going concern. It determines the price at which any buyer can acquire the position. This is what preserves price discovery and enables exit.

Concrete example: **Using our running numbers: a new validator's V^site = 1,000 USDC (equal to F_0, since $N = N_0$ at genesis). They pay 3% × 1,000 = 30 USDC per year in site tax, regardless of network activity. If the network earns well, they might declare V^service = 3,000 USDC. If the network is quiet, they might declare V^service = 500 USDC. The site tax is the same either way. The declared V^service only affects what they receive if someone buys them out.**

This decomposition resolves a tension that has dogged every prior attempt at anti-speculative mechanism design. If you tax only earnings, idle holding is free — the NFT

becomes an art object during dormancy. If you tax the full declared value continuously, validators are punished for network conditions outside their control. By separating the two components and taxing each differently, we get: the anti-speculation property from the site tax, and the zero-dormancy-penalty property from the earnings share.

IV. The Mechanism: Five Instruments

The mechanism has five instruments. They work together, but each one serves a distinct purpose. This section describes each in plain English first, then states the formal definition, then shows the running example.

Instrument 1: The Floor Price

In plain English: **The floor price is the DAO's contractual promise to every validator: at the end of your ten-year lease, we will buy your NFT back at this price. The price rises with every year of honest service. It does not depend on how much the network earned.**

The floor price is the mechanism's ethical core. It says: if you showed up and served honestly, you will get out at least this much. The network's commercial success is irrelevant to this promise. A validator who staked through five years of dormancy and five years of activity gets the same floor as one who staked through ten years of boom. Both served. Both earned their floor.

$$F(\theta) = F_0 \cdot (1 + \alpha \cdot \ln(1 + \theta))$$

Floor price = genesis floor × a tenure multiplier that rises with each year served

The logarithm ensures the floor rises quickly in early years (rewarding early commitment) and more slowly in later years (preventing the floor from growing faster than the reserve can support). α controls the steepness.

Concrete example: **With F_0 = 1,000 USDC and α = 0.40: after 1 year the floor is 1,277 USDC. After 5 years it is 1,643 USDC. After 10 years (full lease) it is 1,960 USDC. The floor nearly doubles over a full lease term.**

Two important features. First, the buy-back price is fixed at the F_0 prevailing when the validator staked — not at the higher F_0 that later cohorts may enjoy. This prevents gaming the mechanism by timing entry to maximise the buy-back. Second, validators who reach years 2, 5, and 8 receive small milestone bonuses (funded from the earnings share pool), making the long-run reward more salient to validators who are prone to under-weighting distant payoffs.

Instrument 2: The Site Tax

In plain English: **The site tax is the Georgist ground rent. Every validator pays it every year, regardless of whether the network is active. It is small but continuous. It is what makes idle speculative holding costly.**

The analogy: **This is Henry George's land tax applied to network positions. Just as land value comes from the community around it, validator position value comes from the network around it. The DAO charges rent for occupying that position, just as a state charges ground rent on land.**

$$T_s(t) = \tau_s \cdot V^{site}(t) = \tau_s \cdot F_0 \cdot (N_0 / N(t))^{\xi}$$

Annual site tax = tax rate × site value, where site value falls as more validators join

The site value V^site falls as more validators join (N increases) because the position becomes less scarce. This means the site tax is highest when the network is smallest and most valuable to be part of, and lowest when the network is large and many validators share the commons.

Concrete example: **With τ_s = 3%, F_0 = 1,000 USDC, N_0 = 20, and ξ = 0.5: at genesis (N = 20) the site tax is 3% × 1,000 = 30 USDC per year. If N grows to 80 validators, site value halves and the site tax falls to 15 USDC per year. If N falls to 5, site value doubles and the tax rises to 60 USDC per year. In all cases, the tax is continuous — a speculator holding an idle position pays it every year.**

The site tax rate τ_s is bounded below by a constitutional minimum τ_{s_min} that cannot be changed by governance. This prevents a captured DAO from zeroing out the anti-speculation property.

Instrument 3: The Earnings Share

In plain English: When the network is active and validators are earning fees, each validator shares a portion of their earnings with the DAO. Longer-serving validators share less, as a reward for loyalty. When the network earns nothing, this share is automatically zero — validators are not taxed on earnings they do not have.

The earnings share rate decreases with tenure. A new validator pays τ_e_0 = 12% of their earnings. A validator who has served for ten years pays approximately 5%, because they have earned the right to a larger share through sustained service.

$$\tau\_e(\theta) = \tau\_e_0 / (1 + \lambda\theta)$$

Earnings share rate = base rate, falling with each year of tenure θ

How much each validator earns from the fee pool depends on their seniority — senior validators are more likely to be selected for tasks and receive a larger share. The seniority function S(θ) = 1 + γ·ln(1+θ) gives this weight. A validator with five years of tenure has S(5) = 1 + γ·ln(6), while a new validator has S(0) = 1.

Concrete example: With τ_e_0 = 12%, λ = 0.20: a new validator pays 12% of their earnings. A 5-year validator pays 12%/(1+0.2×5) = 6%. A 10-year validator pays 12%/(1+0.2×10) = 4%. If the network earns 10,000 USDC in fees and the DAO retains 18%, the 8,200 USDC fee pool is distributed across validators weighted by seniority. A 10-year senior validator might earn 600 USDC that year and pay 24 USDC earnings share. A new validator might earn 200 USDC and pay 24 USDC earnings share.

Instrument 4: The Supply Rule

In plain English: The DAO issues new validator NFTs when the network is near capacity (too much demand for current validators to handle) and retires NFTs when the network is far underutilised. The rate of new issuance is bounded by how quickly the reserve can absorb new buy-back obligations.

The supply rule is governed by the utilisation rate u(t) — the ratio of network demand to total validator capacity. The mechanism defines four regimes:

Regime 0 — Complete dormancy (u = 0): No earnings, no new issuance, no retirements. Site tax continues. Buy-backs are deferred until the network shows signs of life. Validators wait with their floor intact.

Regime 1 — Normal operation (u between thresholds): Steady state. All three income streams active. No supply change needed.

Regime 2 — Near capacity (u approaching 1): New validators are needed. The DAO issues new NFTs, but only as fast as the reserve can back the new buy-back obligations.

Regime 3 — Over capacity (u > 1): Demand exceeds what current validators can handle. Emergency issuance at a lower floor and shorter lease, capped at $N_0/2$ per year.

In plain English: **The growth speed limit is the key safety constraint: the DAO cannot issue new validator NFTs faster than it is earning enough to eventually buy them back. This prevents the network from over-promising during a boom.**

$$\Delta N\_max(t) = DAO\_income(t) / (F_0 \cdot r_0)$$

Maximum new validators per year = DAO annual income divided by the cost of backing one new buy-back obligation

Buy-backs are deferred when the network is below a minimum activity threshold u_buyback. The DAO's obligation is not cancelled — it is queued and paid out gradually when the network recovers, at a rate capped at ψ × DAO_income to prevent a cliff event.

Instrument 5: The DAO Treasury

In plain English: **The treasury accumulates income from three streams — the site tax (always running), the earnings share (running when the network is active), and the DAO's direct fee retention. It deploys these funds to honour buy-back obligations, fund governance, and earn yield on idle reserves.**

$$S(t+1) = S(t) + \rho \cdot R(t) + site\_tax(t) + earnings\_shares(t) - buy\_backs(t) - costs(t) + yield(t)$$

Next year's treasury = this year's treasury + all income − all obligations + reserve yield

The reserve is not left idle. During dormancy periods, the genesis reserve is deployed in a low-risk yield-bearing instrument, earning 3–5% annually. This meaningfully extends the network's ability to honour buy-backs during a prolonged quiet period.

Concrete example: With F_0 = 1,000 USDC, N_0 = 20, and r_0 = 2.0: the genesis reserve is 2.0 × 1,000 × 20 = 40,000 USDC. At 4% annual yield during dormancy, this earns 1,600 USDC per year. The annual buy-back obligation (if all leases were expiring simultaneously at full floor) would be 20 × 1,960 = 39,200 USDC — but leases are staggered, so the actual annual obligation is approximately 3,920 USDC. The reserve comfortably covers this even with zero network revenue.

V. The Four Theorems

This section presents the four core results. Each theorem is preceded by a plain English statement of what it says and why it matters. The formal statement and proof follow for readers who want to verify the result precisely.

Theorem 1: The DAO Is Always Solvent (If You Set the Reserve Correctly)

In plain English: If you raise enough capital at genesis to set the reserve to r_0 times the total initial floor value, the DAO can honour every single buy-back promise even if the network earns absolutely nothing for the entire ten years. The minimum r_0 is a single formula involving only α and the lease length.

This is the most important result in the paper. It transforms the question 'will the DAO be able to pay validators at the end of their lease?' from an actuarial uncertainty into a solved problem. You set the reserve correctly at genesis and the answer is always yes, regardless of what happens to the network.

Concrete example: With α = 0.40 and T = 10 years: r_0_min = 1 + 0.40 × ln(11) = 1.960. The minimum genesis reserve is 1.960 × F_0 × N_0. With F_0 = 1,000 USDC and N_0 = 20: you need to raise at least 39,200 USDC into the buy-back reserve at genesis. Every dollar the network earns above zero is additional safety margin.

Theorem 1 (Unconditional Solvency): ***The DAO satisfies $S(t) \geq 0$ for all $t \in [0, T]$ if and only if the genesis reserve multiplier satisfies $r_0 \geq 1 + \alpha \cdot \ln(1 + T\_lease)$.***

Proof. The maximum buy-back obligation per validator is $F(T) = F_0 \cdot (1+\alpha \cdot \ln(1+T))$, achieved when a validator completes a full lease term. In the worst case $R(t) = 0$ for all t (zero network revenue, zero earnings share), the only income above and beyond the reserve is the site tax $T\_s(t) \geq 0$ and reserve yield $Y(t) \geq 0$. Setting G = 0 for the pure worst case, total buy-back liability is at most $N_0 \cdot F_0 \cdot (1+\alpha \cdot \ln(1+T))$. Genesis reserve is $r_0 \cdot F_0 \cdot N_0$. Solvency requires $r_0 \geq 1+\alpha \cdot \ln(1+T)$. Any $R(t) > 0$ or $Y(t) > 0$ strictly relaxes the constraint. ■

Theorem 2: No Group of Senior Validators Can Monopolise Fees

In plain English: There is a maximum seniority weight γ above which a small group of long-tenured validators would capture too large a fraction of the fee pool. Theorem 2 gives the exact formula for this maximum.

Without this bound, the seniority system could calcify into a permanent oligopoly where early validators capture most fees forever, making it unattractive for new validators to join. The anti-oligopoly bound ensures the seniority premium is meaningful but not unlimited.

Concrete example: With N = 20 validators, T = 10 years, and a 50% oligopoly cap (μ_max = 0.5): the maximum γ is (20/2 - 1) / ln(11) = 9 / 2.398 = 3.75. At γ = 1.0 (our recommended value) the mechanism is well within the safe zone.

Theorem 2 (Anti-Oligopoly Bound): ***For any k validators with maximum tenure T_lease, their combined fee share does not exceed μ_max if and only if γ(k_cohort) ≤ (μ_max⁻¹ − 1)·(N−k) / (k·ln(1+T_lease)).***

Proof. Combined fee share of k validators with tenure T equals k·S(T) / (k·S(T) + (N−k)·S(0)). With S(0)=1 and S(T)=1+γ·ln(1+T), setting this ≤ μ_max and solving for γ yields the stated bound. The binding constraint is at k = ⌊N/2⌋. The cohort-decaying γ(k) = γ_0/k ensures this bound tightens across generations. ■

Theorem 3: A Stable Configuration Exists

In plain English: All five instruments can be simultaneously satisfied. There exists a set of parameter values at which the mechanism is internally consistent — no instrument contradicts another. This proves the mechanism is not self-defeating.

This result uses Kakutani's fixed-point theorem — a mathematical tool for proving that a system with many interacting parts has at least one stable configuration. The supply controller in our mechanism is integer-valued (you cannot have 7.3 validators), which requires the Kakutani version of the theorem rather than the simpler Brouwer version used in earlier drafts.

Theorem 3 (Fixed-Point Existence (Kakutani)): ***There exists a parameter tuple satisfying all five instrument conditions simultaneously.***

Proof. The parameter space Θ is compact and convex by construction of the parameter bounds. The supply rule defines a correspondence $\Phi: \Theta \Rightarrow 2^\Theta$ (set-valued because N is integer-constrained). $\Phi(\theta)$ is non-empty for all θ (at least one integer N satisfies the supply rule for any demand level), convex (the feasible set of N values is an interval of integers with convex hull), and the graph of Φ is closed (follows from continuity of all non-integer instrument functions). By Kakutani's fixed-point theorem, Φ has a fixed point $\Phi^* \in \Theta$. ■

Theorem 4: Future Validators Always Get a Meaningful Floor

In plain English: No matter how many generations of validators stake and exit, and no matter how the network performs, the floor price offered to each new cohort never falls below a governance-set minimum. The mechanism cannot spiral into offering worthless guarantees.

This theorem closes what would otherwise be a long-run failure mode: if the network performs poorly for many successive lease cycles, could the floor price shrink toward zero? The answer is no, because the generational update rule includes a hard floor that governance cannot reduce without a supermajority vote.

Theorem 4 (Generational Floor Boundedness): ***The genesis floor price of cohort k is bounded below by F_0_min for all k ≥ 1.***

Proof. The generational update rule is $F_0^{(k+1)} = \max(F_{0}\text{\_min}, F_0^{(k)} \cdot f(\text{reserve}))$. By the max() operator, $F_0^{(k+1)} \geq F_0\text{\_min}$ for all k ≥ 1 regardless of f(reserve). F_0_min is a Tier 1 immutable governance parameter. ■

VI. Governance Architecture

In plain English: The governance structure has three tiers. The constitutional core (Tier 1) is set at deployment and cannot be changed by any governance vote. It protects the mechanism's anti-speculation property and solvency guarantee from a captured DAO. The treaty layer (Tier 2) can be changed but requires a supermajority and a six-month veto window. Operational parameters (Tier 3) can be adjusted by normal governance within bounds set by Tier 1.

The three-tier structure arose from a specific vulnerability identified in the adversarial audit: without it, a supermajority DAO vote could zero out the site tax (eliminating anti-speculation), set the buy-back threshold to 1.0 (permanently deferring all buy-backs), or

issue new cohorts at near-zero floor prices (extracting the reserve). The constitutional tier prevents all of these.

Tier 1 — Immutable at deployment

- Minimum site tax rate τ_s_min (governance cannot eliminate anti-speculation)
- Maximum buy-back deferral threshold (governance cannot permanently defer buy-backs)
- F_0_min (minimum floor for all generations)
- Slashable offence list (immutable enumeration of cryptographically provable violations)
- Genesis issuance schedule (N_0/M validators per month for M months — prevents day-one full issuance)

Tier 2 — Time-locked supermajority (six-month veto window)

- α (tenure floor premium), γ_0 (seniority weight), λ (earnings discount), T_lease (lease term)

Tier 3 — Normal governance, within Tier 1 bounds

- τ_s, τ_e_0, ρ, supply controller thresholds, deferred queue payout cap ψ, Harberger transfer fee τ_transfer

VII. The Utilisation Oracle

In plain English: **Every supply rule decision depends on u(t) — the utilisation rate. If this number can be manipulated, the entire mechanism is compromised. The oracle design ensures that the utilisation signal comes from a source that validators cannot fake: user-side task requests, not validator completion reports.**

The adversarial audit identified a specific attack: incumbent validators near the issuance threshold have a dominant strategy to under-report task completions, keeping u(t) below the issuance trigger and blocking new validators from entering the network. This is only possible if validators control the utilisation signal. We close this attack by computing u(t) from user request logs, not validator attestations.

- User task requests are logged on-chain at submission time, by the user's wallet, not the validator.
- Task completions are verified by a threshold-signed oracle: k-of-N bonded oracle nodes independently verify each completion, where k > 2N/3.
- u(t) is a rolling 90-day average, not a point-in-time annual measurement. Spike-and-exploit attacks require sustaining artificial demand for 90 days rather than a single measurement window.
- All inputs to u(t) are publicly auditable on-chain. Any governance dispute can be resolved by replaying the computation.

VIII. Enforcement: Slashing

In plain English: **Minor service failures are handled by a continuous performance score that reduces earnings proportionally. Provably malicious acts trigger a three-strike confiscation rule. Slashing is net-positive for the DAO: confiscated value flows to the reserve, directly benefiting all honest validators.**

The performance score $P_i(t) \in [0,1]$ decays with service failures and recovers with honest service. A validator at $P = 0.8$ earns 80% of their seniority-weighted fee allocation. This avoids the bluntness of binary slashing for minor deviations while preserving strong incentives for consistent quality.

The three-strike rule applies only to acts on the immutable Tier 1 offence list: double-signing, Sybil attestation, oracle collusion, reserve theft. These are cryptographically provable and objectively verifiable.

- First strike: $V^{service}$ reduced by 50%. Validator retains position and floor credit.
- Second strike: $V^{service}$ reduced to zero. Cannot sell the position but retains floor credit.
- Third strike: full confiscation. NFT returned to DAO. Tenure and floor credit forfeited. Confiscated value increases the reserve for all honest validators.

IX. How the Mechanism Handles Extreme Scenarios

9.1 Total dormancy — what if the network earns nothing for years?

This is the scenario that most concerns potential validators at genesis. The answer is clear and proven.

- The site tax continues. Validators pay their ground rent. This is small but it prevents free speculative waiting.
- The earnings share drops to zero automatically. No earnings means no earnings tax.
- Buy-backs are deferred, not cancelled. When the network eventually recovers above $u_{buyback}$, the deferred queue pays out at a sustainable rate capped at $\psi \times DAO\_income$ per year.
- The reserve earns yield during dormancy, partially offsetting its draw-down.
- Theorem 1 guarantees the reserve covers all buy-backs even if dormancy lasts the full ten years.

Concrete example: With our running numbers and a five-year dormancy: the reserve of 40,000 USDC earns roughly 4% × 40,000 = 1,600 USDC per year in yield. Annual site tax income is approximately 20 × 30 = 600 USDC. Annual deferred buy-backs queue at a rate of roughly 2 validators per year × average floor ≈ 2,800 USDC. The reserve covers this comfortably and still has approximately 26,000 USDC remaining when the network recovers.

9.2 Explosive growth — what if demand far outstrips capacity?

The growth speed limit prevents the mechanism from over-promising during a boom. New validators can only be issued as fast as the reserve can back their eventual buy-backs. During Regime 3 (demand exceeds capacity), emergency NFTs at the minimum floor and a five-year lease allow rapid capacity response without overcommitting the main reserve.

9.3 Generations — what happens after the first cohort exits?

When the first cohort's leases expire, the DAO uses the accumulated reserve (genesis reserve plus ten years of income minus buy-back payments) to set the floor price for the second cohort. If the network has been active, the second cohort gets a higher floor. If the network has been mostly dormant, the second cohort gets the genesis floor — never less, by Theorem 4. This generational structure means the mechanism compounds upward with success without requiring the network to make promises it cannot keep.

X. Parameter Reference

All parameters with their recommended starting values. Parameters marked [T1] are Tier 1 immutable. Parameters marked [T2] require time-locked supermajority. Parameters marked [T3] are normal governance. Parameters marked [Protocol] require the engineering specification.

- **F_0 — genesis floor price** [T3] The buy-back guarantee for the first cohort. All economic results scale from this.
- **$r_0 \geq 1.960$ — genesis reserve multiplier** [T1] Proven minimum for α=0.40, T=10. Must satisfy $r_0 \geq 1 + \alpha \cdot \ln(1+T)$.
- **α = 0.40 — tenure floor premium** [T2] Controls how fast the floor rises with tenure. 0.30–0.50 recommended.
- **τ_s = 3% — site tax rate** [T3] Annual tax on V^site. Constitutional floor τ_s_min = 1%.
- **τ_e_0 = 12% — base earnings share** [T3] Falls with tenure via λ.
- **λ = 0.20 — tenure earnings discount** [T2] At this value a 10-year validator pays 4.6% earnings share.
- **γ_0 = 1.0 — seniority weight** [T2] Decays across cohorts as γ_0/k. Well within anti-oligopoly bound.
- **ρ = 18% — DAO fee retention** [T3] Share of network fees retained directly by DAO.
- **u_active = 0.05, u_buyback = 0.15** [T3] Earnings activate at 5% utilisation; buy-backs at 15%.

- **u_min = 0.20, u_max = 0.80** [T3] Supply controller dead band.
- **ψ = 0.40 — deferred queue cap** [T3] Maximum fraction of annual DAO income used for deferred buy-back queue.
- **τ_transfer = 2% — Harberger transfer fee** [T3] Creates accurate self-assessment incentive under loss aversion.
- **κ, p_unit, F_0_min** [Protocol] Three fill-ins from engineering specification.

XI. Implementation Notes

The mechanism maps directly onto the existing InfiniteZero smart contract architecture. Seven contracts handle the distinct functions: NFTMinting.sol (tenure tracking, resets on transfer), StakingAndTax.sol (site tax and earnings share), FloorAndReserve.sol (buy-back obligations and deferred queue), SupplyController.sol (regime detection and issuance), PerformanceScore.sol (continuous slashing), OracleConsensus.sol (threshold-signed u(t) computation), and DAO.sol (three-tier governance).

All mechanism values are denominated in stablecoins (USDC or DAI) rather than ETH. This eliminates the mismatch between validator real costs — hardware, bandwidth, time — and a volatile ETH-denominated guarantee. The protocol operates on Ethereum; the accounting is stablecoin-denominated.

Tenure is stored as a signed attestation on Ethereum mainnet, readable by any L2 deployment. If the protocol migrates chains, tenure records survive the migration because they are chain-agnostic.

Genesis validators are onboarded in tranches of N_0/M per month over M months. This distributes buy-back obligations across a range of expiry years rather than concentrating them all at year ten. M is a Tier 1 immutable parameter set at deployment.

XII. Open Problems

This section is not a list of things that are wrong. It is an honest description of the research frontier. The four problems below do not undermine the four proven theorems. They are questions about how the mechanism behaves in practice that the current mathematics cannot fully answer. Each one is a genuine research problem — interesting in its own right, independent of this application.

Open problem 1: Which equilibrium does the system converge to? We prove two equilibria exist — a high-participation thriving network and a low-participation quiet one. We believe the genesis reserve acts as a coordination device that selects the good equilibrium, but this has not been formally proven. A dynamic stability analysis is needed. If the proof fails, we need an additional instrument. Either result is informative.

Open problem 2: Is a quality trap possible? If validators earn too little for too long, they may provide lower-quality service, which reduces demand, which reduces earnings further. This feedback loop could in principle create a third equilibrium — a quality trap — that the mechanism does not automatically escape. We have not ruled this out. The quality production function Q(t) needs to be specified from first principles and its stability verified.

Open problem 3: Is the oracle incentive-compatible? We have designed the oracle to make truthful reporting the rational choice, but we have not formally proven that truthful reporting is a dominant strategy under all conditions. The bonding design is intended to make false attestation costly enough to deter deviation, but the formal proof has not been produced.

Open problem 4: Is the slashing rule renegotiation-proof? After a slash event, the slashed validator could in principle offer a side payment to enforcing validators to reverse the slash. Whether the smart contract architecture prevents this definitively — by making the on-chain commitment more valuable than any off-chain deal — has not been formally shown.

The research programme to address these problems is described in the companion hiring materials document. The programme involves three theoretical collaborators working over 12 months, with the goal of a complete, submission-ready paper by Q1 2027.

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